MARIS — Business Requirements Document (BRD) v1.0

Status: Draft for review
Product: MARIS
Positioning: Wildlife Crime Intelligence & Rapid Response Platform
Development direction: MERN + React Native
Current stage: BRD only — **frontend development does not start until this BRD is approved.**

I have used the uploaded MARIS/SIH and MarineGuard feature documents as the source basis. Where your latest instruction overrides an older technical proposal, I have marked it as a **team decision** rather than pretending the source documents originally specified it. For example, the source architecture proposed PostgreSQL/PostGIS and NestJS, while **you have now explicitly selected MongoDB + MERN + React Native**.

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1. Executive Summary

MARIS is an offline-first wildlife-crime intelligence and rapid-response platform designed to convert fragmented wildlife-crime information into structured, traceable, explainable intelligence for authorized personnel.

The system addresses the gap between:

> **Information → Intelligence → Action**

A citizen, confidential tipster, or field officer may possess a photograph, video, location, time, description, or other evidence. Conventional channels can leave this information fragmented across calls, messages, social media, email, or personal contacts. Connectivity limitations can further prevent timely reporting.

MARIS connects the complete workflow:

> **REPORT → PROTECT → TRANSMIT → INTELLIGENCE → PRIORITIZE → VERIFY → COORDINATE → RESPOND → RECORD**

The source material explicitly distinguishes MARIS from a conventional reporting form: the workflow continues after submission into control-room review, prioritization, officer coordination and response tracking.

The intelligence layer is the core differentiator.

A new report should be capable of being:

- captured offline,
- synchronized later,
- enriched with available intelligence,

- compared with historical incidents,
- matched against geographic/corridor relationships,
- assigned an explainable priority signal,
- escalated when AI is uncertain,
- reviewed by an authorized human,
- and preserved as a traceable case timeline.

The system **supports decisions; it does not make legal or enforcement decisions automatically**. The source material specifically requires human-in-the-loop decision support and confidence-aware AI.

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2. Product Vision

> **MARIS turns field evidence and public information into connected, explainable wildlife-crime intelligence that helps authorized personnel decide what deserves attention and what should be verified.**

Product philosophy

> **Capture the evidence.**
> **Connect it to what is already known.**
> **Explain why it matters.**
> **Help the right human act.**

This is consistent with the core MarineGuard intelligence narrative in the source materials.

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3. Product Positioning

MARIS is not:

- another reporting form,
- merely a chatbot,
- merely an AI species classifier,
- merely a GIS map,
- merely a case-management system,
- an automatic crime detector,
- an automatic legal-decision system.

MARIS is:

> **An intelligence layer between information and action.**

The project materials explicitly emphasize that the strongest concept is not adding disconnected features but making the evidence → intelligence → decision-support chain unusually strong.

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4. Problem Statement

Wildlife-crime information may exist in multiple disconnected forms:

- field observations,
- photographs/videos,
- citizen reports,
- confidential tips,
- incident reports,
- historical records,
- geographic information,
- species information,
- commodity/trade information,
- enforcement records.

A person may possess valuable evidence but lack a reliable pathway to transmit it. Organizations then have to determine:

- whether the report is useful,
- whether it is urgent,
- whether it relates to previous activity,
- whether evidence is sufficient,
- who should verify it,
- and what response is appropriate.

The source documents identify connectivity failure, fragmented information, source-exposure concerns, inconsistent evidence, multiple simultaneous incidents and the lack of a reliable intelligence-to-response workflow as major gaps.

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5. Business Objectives

MARIS shall aim to:

BO-01 — Increase reportability

Allow important information and evidence to be captured even when internet connectivity is unavailable.

BO-02 — Improve evidence quality

Structure reports around:

Photo/Video + Location + Time + Incident Details

rather than unstructured messages.

BO-03 — Reduce information fragmentation

Bring citizen reports, field reports, evidence and historical incidents into a shared intelligence layer.

BO-04 — Improve intelligence discovery

Identify relevant relationships between new incidents and available historical information.

BO-05 — Improve prioritization

Provide explainable priority signals so authorized personnel can focus attention appropriately.

BO-06 — Preserve uncertainty

Prevent AI from presenting uncertain identification or analytical conclusions as facts.

BO-07 — Improve coordination

Connect the control room with appropriate authorized field personnel.

BO-08 — Improve institutional memory

Preserve incident, evidence, verification and response history.

These objectives align with the documented expected benefits: centralized intelligence, better evidence/context, faster prioritization, multiple-incident management, officer coordination and response tracking.

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6. Target Users

MARIS will support the following user groups.

6.1 Citizen

A member of the public who observes suspected wildlife-crime activity.

Needs

- simple reporting,
- minimal friction,
- evidence upload,
- location,
- description,

- confidentiality,
- limited status visibility.

The source material describes a lightweight citizen reporting interface rather than duplicating the full authority application.

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7. Confidential Tipster

A person who possesses sensitive information and is concerned about identity exposure.

Needs

- pseudonymous reporting,
- minimal identity exposure,
- evidence submission,
- controlled communication,
- restricted access,
- report tracking where supported.

MARIS shall distinguish:

> **Tip ≠ Proof ≠ Confirmed Crime**

A tip may be important without being verified.

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8. Field Officer

A primary MARIS user.

Needs

- rapid field capture,
- offline operation,
- camera,
- GPS,
- timestamp,
- species/product identification,
- multi-item incident logging,
- evidence management,
- historical matching,
- intelligence signals,
- verification workflow,
- response updates.

The authority-side application is explicitly identified as the highest-priority interface in the feature scope.

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9. Control Room Operator

Responsible for managing incoming information and coordinating cases.

Needs

- incoming reports,
- priority queue,
- intelligence context,
- evidence review,
- map/location,
- assignment,
- escalation,
- multiple active cases,
- officer coordination,
- response tracking.

The control-room concept is central to the MARIS workflow.

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10. Supervisor

Provides operational oversight.

Responsibilities

- review important cases,
- monitor case workload,
- oversee assignments,
- review verification,
- approve/escalate where authorized,
- inspect case history,
- monitor analytical decisions.

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11. Organization Administrator

Manages an authorized organization's MARIS environment.

Responsibilities

- users,
- roles,
- permissions,
- organization settings,
- authorized operational areas,
- access control,
- audit visibility.

The source documents explicitly mention controlled organizational onboarding and role-based access.

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12. Product Interfaces

MARIS will have **three primary user-facing experiences**.

Interface A — Authority Mobile App

Technology: React Native

Primary users:

- Field Officer
- Supervisor
- authorized field personnel

Primary purpose:

> **Capture → Work Offline → Synchronize → Investigate → Verify → Respond**

Interface B — Citizen/Tipster Reporting

Technology: React web / responsive experience initially.

Primary purpose:

> **Photo → Location → Description → Submit → Status**

The source material specifically recommends a lightweight responsive citizen experience rather than reproducing the full offline authority application.

Interface C — Control Room Web Application

Technology: React

Primary users:

- Control Room Operator
- Supervisor
- Organization Administrator

Primary purpose:

> **Receive → Understand → Prioritize → Assign → Coordinate → Track**

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13. Core Product Workflow

The canonical MARIS workflow shall be:

OBSERVE
↓
CAPTURE
↓
PROTECT
↓
STORE / SYNC
↓
IDENTIFY
↓
CONNECT
↓
EXPLAIN
↓
PRIORITIZE
↓
VERIFY
↓
COORDINATE
↓
RESPOND
↓
RECORD

The documented core intelligence loop is:

> **CAPTURE → PRESERVE → SYNC → IDENTIFY → CONNECT → EXPLAIN → TRACK → VERIFY**

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14. Functional Requirements

FR-01 — Incident Creation

MARIS shall allow an authorized user to create an incident.

Minimum information:

- incident ID,
- incident type,
- date/time,
- location,
- description,
- source type,
- evidence,
- status.

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15. FR-02 — Evidence Capture

The authority application shall support:

- photograph capture,
- video capture,
- existing media selection,
- timestamp,
- GPS/location,
- structured notes.

The source material identifies photo/video, GPS, timestamp and incident details as the core evidence package.

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16. FR-03 — Offline Incident Creation

The authority application shall remain functional for core incident creation without internet access.

Offline functionality shall include:

- create incident,
- capture evidence,
- record location,
- record time,
- add items,
- save report,
- edit unsynchronized report,
- view sync status.

17. FR-04 — Offline Queue

When connectivity is unavailable:

Incident
↓
Local Storage
↓
Pending Sync

The user shall see an explicit state such as:

> **Saved securely on device — waiting for connection**

This is directly aligned with the intended airplane-mode demonstration.

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18. FR-05 — Automatic Synchronization

When connectivity returns:

Pending
↓
Uploading
↓
Server Received
↓
Processed
↓
Synced

The system shall:

- synchronize automatically,
- preserve report identity,
- avoid duplicate creation,
- retry failed synchronization,
- retain synchronization state,
- report failures clearly.

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19. FR-06 — Multi-Item Incident

One incident shall support multiple items.

Example:

CASE #MARIS-00124

Item 1
Species: ...
Commodity: ...
Quantity: ...

Item 2
Species: ...
Commodity: ...
Quantity: ...

Item 3
Species: ...
Commodity: ...
Quantity: ...

This is a P0 requirement because the project specifically identifies real-world seizures as potentially containing multiple species/commodities.

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20. FR-07 — Species/Product Identification

MARIS shall support AI-assisted identification.

Output shall use language such as:

> **Possible identification**

rather than:

> **Confirmed identification**

The source material explicitly prohibits universal/100% identification claims.

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21. FR-08 — Confidence-Aware AI

Every AI identification result shall have a confidence state.

Possible states:

- High confidence

- Medium confidence
- Low confidence
- Unknown

If confidence falls below the configured operational threshold:

> **Identification uncertain — human verification recommended.**

The user shall be able to continue the case using:

> **Unknown / Unidentified**

instead of being forced to select an incorrect label.

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22. FR-09 — Historical Incident Matching

MARIS shall compare new incidents with available historical records.

Potential matching dimensions:

- species,
- commodity,
- location,
- route/corridor,
- time window,
- quantity,
- incident type,
- other available evidence.

The result should answer:

> **Have we seen something like this before?**

The project explicitly identifies live historical/corridor matching as one of the highest-leverage capabilities.

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23. FR-10 — Pattern Relationships

The system may show:

> **3 previous incidents match this species + route/corridor.**

It shall provide:

- supporting incidents,
- matching attributes,

- locations,
- dates,
- source information where available.

It shall **not** state:

> Criminal network identified.

unless independently established by authorized human investigation.

The source explicitly requires recurring relationships to be described as relationships in available evidence rather than automatic criminal attribution.

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24. FR-11 — Explainable Priority

MARIS shall generate an evidence-based priority signal.

Possible factors:

- species significance,
- historical activity,
- geographic signal,
- commodity signal,
- quantity anomaly,
- recency,
- route association.

The exact weights shall be treated as implementation parameters and shall not be presented as validated until actually implemented and tested.

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25. FR-12 — Separate Priority, Confidence and Evidence Strength

These are separate attributes.

Example:

Priority
HIGH — 84/100

AI Confidence
74%

Evidence Strength

MEDIUM

Verification
PENDING

This prevents a high AI confidence from being interpreted as proof of illegal activity.

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26. FR-13 — Why Was This Flagged?

Every analytical priority signal shall expose its contributing evidence.

Example:

WHY WAS THIS FLAGGED?

+22 Same species as previous incidents
+19 Location overlaps historical activity
+16 Commodity matches historical records
+14 Recent relevant activity
+08 Quantity anomaly

Priority: HIGH

Supporting records shall be accessible.

The source material explicitly states that judges/users should not be asked to trust a black box; the evidence behind the signal should be visible.

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27. FR-14 — Human Verification

MARIS shall provide a human verification workflow.

Possible outcomes:

- Verified
- Not verified
- Incorrect identification
- Insufficient evidence
- Requires further investigation
- Uncertain

The system shall preserve:

- verifier,

- timestamp,
- decision,
- notes,
- supporting evidence.

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28. FR-15 — Decision Support

MARIS may recommend verification steps.

Example:

> **Suggested verification**
>
> - Confirm species identity.
> - Compare commodity with historical incidents.
> - Review location relationship.
> - Inspect original evidence.
> - Confirm quantity.

The system shall not automatically determine:

- guilt,
- legality,
- criminal liability,
- enforcement action.

The final decision belongs to authorized personnel.

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29. FR-16 — Control Room

The control room shall provide:

- incoming reports,
- case list,
- priority,
- status,
- evidence preview,
- location,
- intelligence signals,
- assignment,
- escalation,
- response tracking.

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30. FR-17 — Priority Queue

Cases should be sortable/filterable by:

- priority,
- status,
- location,
- date,
- incident type,
- assigned officer,
- verification state.

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31. FR-18 — Multiple Active Incidents

The control room shall allow multiple cases to remain active simultaneously.

One case shall not block the processing of another.

This is explicitly identified as a requirement in the MARIS operational model.

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32. FR-19 — Officer Assignment

Authorized control-room users shall be able to assign cases to appropriate field officers.

Assignment information shall include:

- officer,
- assignment time,
- case,
- location,
- priority,
- current status.

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33. FR-20 — Response Tracking

Field officers shall be able to update:

- assignment accepted,
- travelling,
- on site,

- evidence collected,
- verification performed,
- response completed.

The exact operational statuses can be finalized during system design without changing the BRD requirement.

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34. FR-21 — Case Timeline

Every important case event shall be recorded.

Core timeline events:

- evidence captured,
- location recorded,
- incident created,
- saved offline,
- synchronized,
- AI identification performed,
- confidence recorded,
- historical matches generated,
- priority assigned,
- reason generated,
- assignment,
- verification,
- status change,
- closure.

The Feature Master explicitly defines these events.

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35. FR-22 — Evidence Provenance

For each evidence object, MARIS shall preserve available metadata including:

- evidence ID,
- original reference,
- timestamp,
- location,
- capture/source information,
- synchronization state,
- associated incident,
- analytical events referencing it.

The system shall **not claim legal or forensic proof** merely because evidence metadata is preserved.

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36. FR-23 — Citizen Reporting

Citizen reporting shall support:

- photo,
- location,
- short description,
- optional additional information,
- pseudonymous/anonymous pathway where appropriate,
- submission confirmation,
- basic status.

The citizen interface should remain substantially simpler than the authority interface.

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37. FR-24 — Tipster Protection

MARIS shall minimize unnecessary exposure of tipster identity.

A pseudonymous identifier shall be used for operational interaction.

Proposed identity structure

Pseudonymous Tipster ID
    +
Report ID
    +
Internal protected identity record, if identity is ever collected

The operational interface should not require control-room staff to see personal identity unless their role explicitly authorizes it.

The source material specifies a 10-digit pseudonymous Tipster ID.

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38. FR-25 — Tip ≠ Proof

A submitted tip shall remain distinguishable from:

- evidence,
- AI output,
- verified information,
- confirmed incident.

MARIS shall therefore maintain separate fields for:

> **Source type / Evidence / Verification / Analytical signal**

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39. FR-26 — BiChat

Based on your clarification:

> **BiChat is an application/channel intended for communication under low internet connectivity.**

MARIS shall treat BiChat as a communication integration rather than inventing an underlying transport technology that has not been specified.

The BRD therefore defines:

Required capability

- low-connectivity communication,
- connection state,
- message state,
- controlled participants,
- relevant case/report association where applicable.

Not yet specified

The exact technical transport mechanism.

This will be defined during technical design after the actual BiChat technology is confirmed.

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40. FR-27 — Search

Authorized users shall be able to search available records.

Search targets may include:

- incident ID,
- species,
- commodity,
- location,
- date,
- case status,
- source,
- officer,

- report type.

Similar-incident search is identified as P1/P2 rather than one of the absolute P0 features.

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41. FR-28 — Basic Geospatial Visualization

MARIS shall provide basic incident-location visualization.

At minimum:

- incident location,
- current case,
- related incidents,
- filtering.

Large graph/network visualization is deliberately excluded from the initial scope.

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42. Role-Based Access Control

Citizen

Can:

- create report,
- view own limited report information.

Cannot:

- view internal intelligence,
- view other reports,
- view sensitive case information,
- modify organizational decisions.

Tipster

Can:

- submit information,
- view limited report state,
- communicate through approved confidential mechanism if implemented.

Cannot:

- view internal investigation information.

Field Officer

Can:

- create incidents,
- view assigned cases,
- capture evidence,
- update assigned cases,
- perform authorized verification,
- view relevant intelligence.

Cannot:

- manage organization-wide users unless separately authorized.

Control Room Operator

Can:

- view incoming reports,
- review evidence,
- prioritize,
- assign,
- coordinate,
- update operational status.

Supervisor

Can:

- oversee cases,
- review assignments,
- review verification,
- escalate,
- monitor operations.

Organization Administrator

Can:

- manage users,
- manage roles,
- manage organizational settings,
- manage permissions.

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43. Security Requirements

MARIS shall follow security-by-design principles.

Security requirements

- authenticated staff access,
- role-based authorization,
- protected sessions,
- encrypted network communication,
- protected sensitive evidence,
- controlled access to tipster identity,
- audit logging,
- data minimization,
- secure local storage,
- controlled synchronization,
- secure file access.

The project material explicitly identifies role-based access, pseudonymous identity, encryption and controlled access as security requirements.

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44. Privacy Requirements

MARIS shall follow:

> **Minimum necessary information principle**

If personal identity is not required for operational processing, it should not be unnecessarily exposed.

Sensitive locally stored information may be deleted after successful transmission according to the eventual retention policy, while the authorized incident record may remain operationally available. This behavior is described in the submitted concept.

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45. Offline Synchronization Requirements

This is one of the most important engineering areas.

Local states

DRAFT
↓
READY_TO_SYNC
↓
SYNCING
↓
SYNCED

Failure states:

SYNC_FAILED
CONFLICT

The UI must make these states understandable.

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46. Duplicate Prevention

A report created offline must retain a unique client-side identifier.

When synchronized:

Client ID
↓
Server checks existing record
↓
Already exists? → update/acknowledge
↓
Doesn't exist? → create

The system must not create two incidents because the user pressed retry.

The source's live-sync requirement specifically calls for synchronization without duplicate records.

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47. Data Model — Business Level

The BRD requires at least the following conceptual entities.

Organization
User
Role

Permission
Tipster
Incident
IncidentItem
Evidence
Location
AIAnalysis
HistoricalMatch
PrioritySignal
Verification
Assignment
Response
CaseTimelineEvent
Notification
Communication
Source
AuditLog

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48. Core Relationship Model

```
Organization
   │
   └── Users
         │
         └── Roles

Tipster
   │
   └── Incidents
         │
         ├── Incident Items
         ├── Evidence
         ├── Location
         ├── AI Analysis
         ├── Historical Matches
         ├── Priority Signals
         ├── Assignments
         ├── Verification
         ├── Response
         └── Timeline Events
```

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49. MongoDB Decision

Team decision — LOCKED

We will use:

MongoDB

instead of the PostgreSQL/PostGIS architecture proposed in an earlier technical document.

The uploaded technical document originally specifies PostgreSQL + PostGIS, while your latest decision explicitly changes the implementation direction to MongoDB + MERN + React Native.

Therefore the BRD will use MongoDB as the authoritative application database.

MongoDB geospatial capabilities will be used for the initial location-based functionality.

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50. Technology Stack

Web

React + TypeScript

Mobile

React Native + TypeScript

Backend

Node.js + Express + TypeScript

Database

MongoDB

Local mobile database

SQLite / equivalent React Native local persistence

Evidence storage

Object/file storage service.

AI/Analytics

Separate service/module as needed.

The source materials originally proposed React Native, Node.js, database storage, GIS, secure authentication and explainable rule/ML analytics; the exact backend/database choices have now been updated according to your explicit team decision.

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51. Architecture

```
┌──────────────────────────────┐
│     CITIZEN        │
│  React Web App      │
└─────────────┬──────────────┘
      │
      │
┌─────────────▼──────────────┐
│  AUTHORITY APP       │
│ React Native        │
│              │
│ Camera           │
│ GPS            │
│ Evidence          │
│ SQLite           │
│ Offline Sync        │
└─────────────┬──────────────┘
      │
      ▼
 ┌──────────────────────┐
 │ Express REST   │
 │ API / Node.js  │
 └──────────┬───────────┘
      │
┌───────────────┼──────────────────┐
▼       ▼       ▼
┌──────────────┐ ┌──────────────────┐ ┌──────────────────┐
│ MongoDB  │  │ File/Object│  │ Intelligence│
│      │ │ Storage   │ │ Services   │
└──────────────┘ └──────────────────┘ └────────┬─────────┘
         │
     ┌─────────────────┴────────────────┐
     │            │
  ML Models       Rule Engine
     │            │
     └────────────────┬─────────────────┘
         ▼
     Explainable Signal
         │
         ▼
     ┌───────────────────────────────┐
```

```
│  CONTROL ROOM WEB     │
│ React + TypeScript    │
└──────────────┬──────────────┘
        │
        ▼
  Authorized Officers
```

---

52. AI Architecture

MARIS will use a **hybrid intelligence approach**.

```
        INCIDENT
           │
   ┌───────────────┴──────────────┐
   ▼           ▼
 RULE ENGINE       ML MODELS
   │           │
   │           │
   └──────────────┬───────────────┘
          ▼
       SIGNAL ENGINE
          │
   ┌──────────────┼──────────────┐
   ▼      ▼      ▼
 Priority  Confidence  Evidence
          │
          ▼
       Explainability
          │
          ▼
      Human Verification
```

Important rule

> **AI output is decision support, not legal determination.**

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53. AI Output Contract

The system should return something conceptually like:

Identification:
Possible Seahorse

Confidence:

81%

Priority:
HIGH — 84/100

Evidence Strength:
MEDIUM

Historical Match:
3 related incidents

Why Flagged:
• Species relationship
• Location relationship
• Commodity relationship
• Recent historical activity

Verification:
Human verification recommended

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54. Uncertainty Policy

MARIS shall never force a clean AI result solely because the interface expects one.

The system must support:

> **Unknown / Unidentified**

and:

> **Human verification recommended**

This is explicitly identified as a core responsible-AI capability.

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55. Case Lifecycle

Internal lifecycle

RECEIVED
↓
SCREENING
↓
PRIORITIZED
↓
ASSIGNED

↓
UNDER VERIFICATION
↓
VERIFIED
↓
ACTIONED
↓
CLOSED

Possible alternate states:

- Duplicate
- Rejected
- On Hold
- Escalated

These states must be implemented in a way that preserves the audit trail.

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56. Citizen-Facing Lifecycle

Citizens should see a simplified state:

SUBMITTED
↓
UNDER REVIEW
↓
ESCALATED / IN PROGRESS
↓
CLOSED

They should not automatically receive:

- officer identity,
- sensitive locations,
- intelligence matches,
- internal priority reasoning,
- confidential evidence.

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57. Evidence Chain

The case timeline is a first-class business object.

Example:

CASE #MARIS-1042

08:42 Evidence captured
08:43 Location recorded
08:45 Incident created
08:45 Saved offline
09:12 Synchronized
09:13 AI identification performed
09:13 Confidence recorded
09:14 Historical matches generated
09:15 Priority assigned
09:15 Why-flagged explanation generated
09:18 Assigned for verification
10:30 Verification completed
11:05 Case updated

This is directly based on the documented Evidence Chain feature.

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58. Non-Functional Requirements

NFR-01 — Offline Reliability

Core authority reporting must remain usable without internet.

NFR-02 — Data Integrity

Offline synchronization must prevent duplicate incidents and preserve identifiers.

NFR-03 — Security

Sensitive information must be protected through authentication, authorization and controlled access.

NFR-04 — Auditability

Important system and user actions must be traceable.

NFR-05 — Explainability

Important AI-derived signals must expose their supporting factors.

NFR-06 — Usability

Field workflows must minimize typing and unnecessary navigation.

The source material specifically identifies field speed, minimal typing and reliability under poor connectivity as authority-app design goals.

NFR-07 — Scalability

The architecture should allow:

- multiple organizations,
- multiple zones,
- multiple officers,
- multiple simultaneous incidents.

NFR-08 — Maintainability

Frontend, backend, synchronization and intelligence components should remain modular.

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59. Priority Classification

P0 — Must Build

These are the core MARIS differentiators:

1. Authority offline-first reporting
2. Evidence capture
3. Multi-item incident logging
4. Offline-to-online synchronization
5. AI-assisted identification
6. Uncertainty handling
7. Historical/corridor matching
8. Explainable "Why Flagged"
9. Human-in-the-loop verification
10. Evidence Chain / Case Timeline

This matches the Feature Master, where the first five capabilities are treated as core and the flagship loop is centered around offline reporting, matching, explainability and uncertainty.

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60. P1 — Important

- Control-room prioritization
- Case assignment
- Response tracking
- Citizen reporting

- Basic map
- Evidence/source traceability
- Search
- Role management
- Tipster status

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61. P2 — Later

- Advanced similar-incident search
- More sophisticated intelligence analytics
- Larger historical intelligence exploration
- Expanded organization management
- Advanced GIS functionality
- More advanced communication integration

The source Feature Master places similar-incident search and priority/case management below the core P0 capabilities.

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62. Explicitly Deprioritized

The current development scope will **not** prioritize:

Crime forecasting as a headline feature

Generic citizen chatbot

Large network/graph dashboard

Unrelated technology added only for novelty

The MarineGuard scope document explicitly recommends deprioritizing these features.

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63. What MARIS Must Never Claim

The UI, AI outputs and presentation shall avoid statements such as:

> AI detected a criminal.
>
> AI proved this shipment is illegal.
>
> Criminal network identified.

AI knows where trafficking will occur.

82% probability this shipment is illegal.

Instead:

Possible identification.

Priority for further investigation.

Recurring relationship in available evidence.

Trend/projection.

Decision support.

Human verification required.

This language policy is explicitly specified in the project source.

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64. Key Demo Acceptance Scenario

The most important end-to-end demonstration shall be:

Step 1

Officer encounters suspected wildlife crime.

Step 2

Device enters airplane mode.

Step 3

Officer captures:

- photograph,
- location,
- quantity,
- incident details.

Step 4

MARIS saves the report locally.

Step 5

UI displays:

Saved securely — waiting for connection

Step 6

Connectivity returns.

Step 7

MARIS automatically synchronizes.

Step 8

System checks historical intelligence.

Step 9

Example:

3 previous incidents match this species + route

Step 10

MARIS displays:

Why was this flagged?

with supporting evidence.

Step 11

An ambiguous image is submitted.

Step 12

MARIS responds:

Identification uncertain — human verification recommended.

Step 13

Authorized officer makes the final decision.

This exact sequence is recommended as the strongest two-minute demonstration in the source material.

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65. Success Criteria

MARIS V1 should be considered successful if it can demonstrate:

Engineering

- report creation offline,
- evidence capture offline,
- reliable synchronization,
- duplicate prevention.

Intelligence

- historical incident matching,
- geographic/corridor relationship,
- structured intelligence output.

Explainability

- visible reasons for priority,
- supporting records.

Responsible AI

- confidence displayed,
- uncertain result handled without forced classification.

Operations

- case assignment,
- verification,
- response status,
- timeline.

Traceability

- evidence provenance,
- analytical events,
- human decisions.

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66. Business Value by Stakeholder

Stakeholder	Current problem	MARIS value
Citizen	Doesn't know where/how to report	Structured reporting pathway
Confidential tipster	Identity exposure concern	Pseudonymous controlled reporting
Field officer	Connectivity + fragmented information	Offline evidence + intelligence
Control room	Multiple reports and prioritization	Centralized intelligence workflow

Supervisor	Limited operational visibility	Case oversight + timeline
Organization	Information spread across channels	Institutional intelligence

These stakeholder benefits are consistent with the submitted impact model.

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67. Key Risks and Mitigation

Risk	MARIS mitigation
No internet	Offline-first storage + sync
Duplicate sync	Client IDs + idempotent synchronization
False report	Human verification
Wrong AI identification	Confidence + uncertainty
Sensitive source exposure	Pseudonymous identity + RBAC
Poor evidence quality	Structured reporting
Multiple incidents	Parallel control-room workflow
AI overreach	Human-in-the-loop
Data inconsistency	Validation + source traceability
Unsupported technical claims	Explicit scope/architecture validation

The source feasibility document identifies connectivity, false reports, sensitive data, simultaneous incidents, unauthorized organizations and poor-quality reports as key risks.

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68. Important BRD Boundary

There are two layers in the product:

Operational Layer

What happened?

- Report
- Evidence
- Location
- Time

- Items
- Assignment
- Response

Intelligence Layer

What might this information mean in context?

- Identification
- Historical match
- Pattern
- Priority
- Explanation
- Uncertainty
- Verification recommendation

This separation is important because **raw evidence and analytical interpretation must never be treated as the same thing**.

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69. Final Product Story

The complete MARIS story is now:

FIELD OBSERVATION
↓
PHOTO / VIDEO / LOCATION / TIME
↓
MARIS
↓
OFFLINE-FIRST PRESERVATION
↓
SECURE SYNCHRONIZATION
↓
INTELLIGENCE ENRICHMENT
↓
SPECIES / PRODUCT IDENTIFICATION
↓
HISTORICAL + CORRIDOR MATCHING
↓
EXPLAINABLE PRIORITY
↓
UNCERTAINTY HANDLING
↓
HUMAN VERIFICATION
↓
CONTROL ROOM

↓
AUTHORIZED OFFICER
↓
FIELD RESPONSE
↓
DIGITAL CASE TIMELINE
↓
INSTITUTIONAL INTELLIGENCE

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70. Final Product Statement

I recommend locking this as the central BRD statement:

> **MARIS transforms wildlife-crime reporting from a simple submission process into a resilient, confidential, evidence-driven and explainable intelligence-to-response workflow.**

And the strongest positioning line remains:

> **“We are not adding another reporting form. We are creating the intelligence layer between information and action.”**